Magnon Bound States in One-Dimensional Heisenberg Chain

Advanced Solid State Physics

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Outline

The Problem

- **Magnons:** Quasiparticles representing propogating spin flips.
- Strong effective interactions between magnons can overcome their kinetic tendency to delocalize.
- **Result:** Formation of stable, localized bound states.

Heisenberg Spin Hamiltonian

$$\hat{H} = -\frac{J}{2}\sum_{i=1}^{N}\left(\sigma_i^x\sigma_{i+1}^x + \sigma_i^y\sigma_{i+1}^y\right) + J_z\sum_{i=1}^{N}\sigma_i^z\sigma_{i+1}^z - h\sum_{i=1}^{N}\sigma_i^z$$

Mapping to Fermions: Jordan-Wigner Transformation

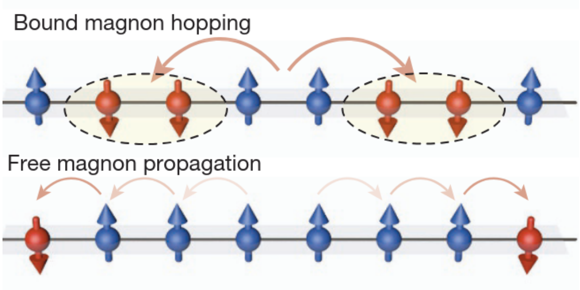
- Spin-1/2 Hamiltonian $\rightarrow$ Interacting Spinless Fermions
- **Physical Equivalencies:**
  - Local spin flip $\rightarrow$ Fermion creation/annihilation
  - Transverse exchange (J) $\rightarrow$ Nearest-neighbor hopping (t)
  - Longitudinal coupling (J_z) $\rightarrow$ Density-density interaction (U)

Mapped Fermionic Hamiltonian

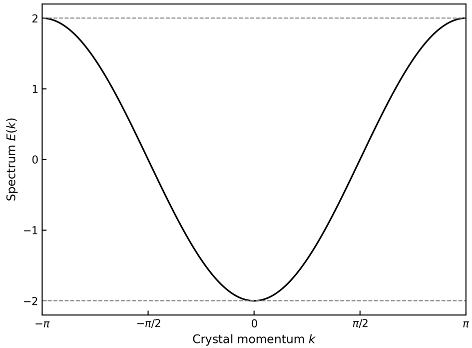
$$\hat{H} = -t \sum_i \left(c_i^\dagger c_{i+1} + \text{h.c.} \right) + U \sum_i n_i n_{i+1}$$

Few Fermions

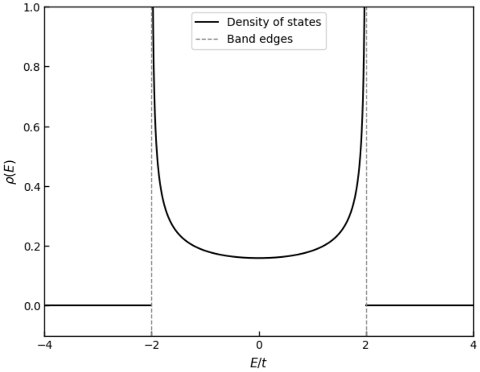
- We focus on the two-particle sector of the fermionic model.
- Equivalent to two spin flips (magnons) in the original spin model.
- Goal: Analyze the formation and dynamics of magnon bound states.



Non-Interacting Limit: 1D Tight-Binding Model

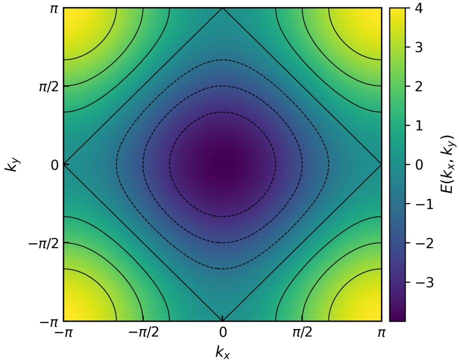


Energy Dispersion $\epsilon(k)$

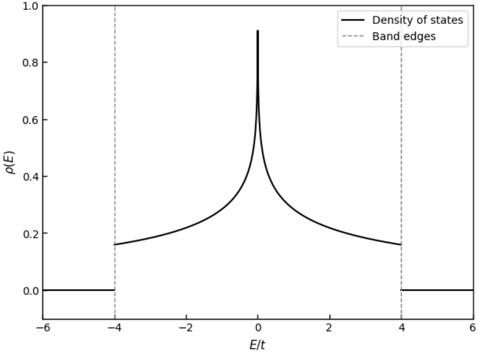


Density of States $\rho(E)$

Non-Interacting Limit: 2D Tight-Binding Model on a Square Lattice



Energy Dispersion $\epsilon(k_x, k_y)$



Density of States $\rho(E)$

Note: The 2-particle non-interacting DOS in 1D mathematically mirrors this 1-particle DOS in 2D.

Resolvent Operator and Green's function

- Two-particle basis:

$$|k,n\rangle = \frac{1}{\sqrt{N}} \sum_j exp\left(ik\left(R_j + \frac{na}{2}\right)\right) c_j^{\dagger} c_{j+n}^{\dagger}|0\rangle \tag{1}$$

- Green's Function from resolvent:

$$G(m,n;k,t) = \langle k,m|\, e^{-iHt}\, |k,n\rangle$$
$$G(m,n;k,\omega) = \langle k,m|\, (\omega + i\eta - \hat{H})^{-1}\, |k,n\rangle$$

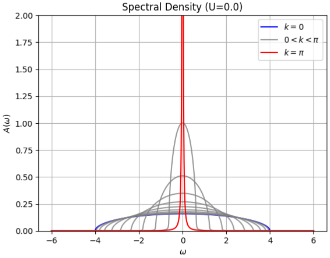
Recurrence Relation:

$$\delta_{m,n} = (\omega + i\eta - U\delta_{n,1})G_{m,n} + f(k)(G_{m,n-1} + G_{m,n+1})$$

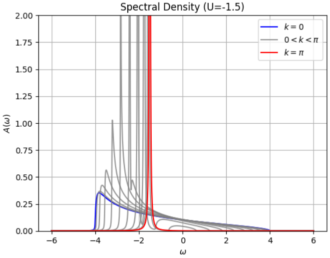
(where $f(k) = 2t\cos(ka/2)$ is the effective hopping of relative motion)

- $G(1,1;k,\omega)$: Bound state to bound state propagator.
- The spectral density of these two-particle states:

$$A_2(k,\omega) = -\frac{1}{\pi}\mathrm{Im}\, G(1,1;k,\omega)$$



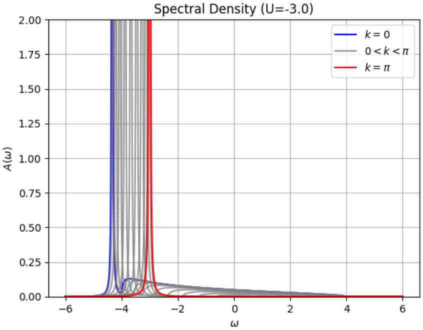
$U = 0$ (Scattering Continuum)



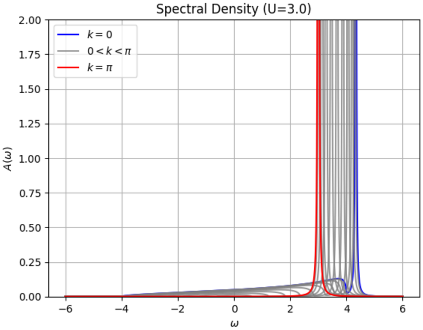
$U = -1.5t$ (Bound state peak forming)

Spectral Weights: Emergence of the Bound State

- For strong interactions ($|U| > 2t$), a discrete pole separates from the continuum.



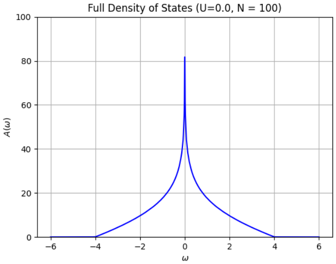
$U = -3t$ (Attractively Bound State)



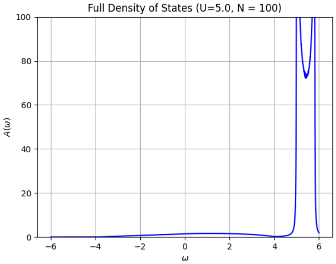
$U = +3t$ (Repulsively Bound State)

Local Density of States (Nearest-Neighbor Sector)

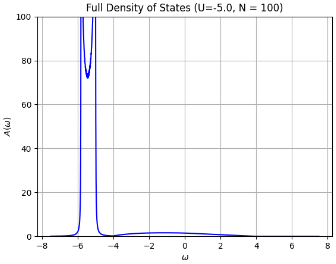
$U = 0$



$U = 5t$



$U = -5t$



$$\rho_{nn}(\omega) = \sum_{k \in \mathrm{BZ}} A_2(k, \omega)$$

What is the Bound State?

The "Doublon"

- Two fermions rigidly localized at nearest-neighbor sites.
- Creation operator: $b_j^\dagger = c_j^\dagger c_{j+1}^\dagger$

Quantum Statistics

- Commutator depends on spacing.
- Behaves like a *Hard-Core Boson*.
- Subject to underlying Pauli exclusion.

$$[b_i, b_j^\dagger] = \delta_{ij}(1 - n_{i+1} - n_i) + \delta_{i+1,j}\ c_{i+2}^\dagger c_i + \delta_{i,j+1}\ c_j^\dagger c_{j+2}$$

Quantum Statistics of the Doublon

$$[b_i, b_j^\dagger] = \delta_{ij}(1 - n_{i+1} - n_i) + \delta_{i+1,j}\ c_{i+2}^\dagger c_i + \delta_{i,j+1}\ c_j^\dagger c_{j+2}$$

1. **Separated Pairs ($|i-j| > 1$)** $\implies$ Pure Boson Behavior
2. **On-Site Overlap ($i = j$)** $\implies$ Hard-Core Boson Behavior

$$|0,0\rangle \xrightarrow{b_i^\dagger} |1,1\rangle \quad |1,0\rangle \xrightarrow{b_i^\dagger} 0 \quad |1,1\rangle \xrightarrow{b_i} |0,0\rangle$$

$$\implies \quad [b_i, b_i^\dagger] = 1 - n_i - n_{i+1}$$

3. **Nearest-Neighbor ($j = i+1$)** $\implies$ Effective Hopping

$$|1,0,0\rangle \xrightarrow{b_{i+1}^\dagger} |1,1,1\rangle \xrightarrow{b_i} |0,0,1\rangle \implies \quad [b_i, b_{i+1}^\dagger] = c_{i+2}^\dagger c_i$$

Effective Hamiltonian: Feshbach Projection Formalism

Strong Coupling Regime ($|U| \gg t$):

- Project dynamics purely into bound-state subspace (P-space).
- Integrate out the orthogonal subspace (Q-space).

$$\begin{pmatrix} H_{PP} & H_{PQ} \\ H_{QP} & H_{QQ} \end{pmatrix} \xrightarrow{\text{Resolvent}} \begin{pmatrix} \omega - H_{PP} & -H_{PQ} \\ -H_{QP} & \omega - H_{QQ} \end{pmatrix}^{-1} = \begin{pmatrix} G_{PP} & G_{PQ} \\ G_{QP} & G_{QQ} \end{pmatrix}$$

$$G_{PP} = (\omega - H_{PP} - H_{PQ}(\omega - H_{QQ})^{-1} H_{QP})^{-1} = (\omega - H_{\text{eff}})^{-1}$$
$$\implies H_{\text{eff}} = H_{PP} + H_{PQ}(\omega - H_{QQ})^{-1} H_{QP}$$

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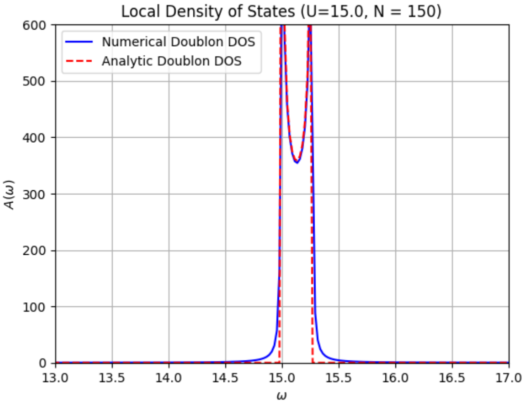
Effective Hamiltonian for the Doublon ($\omega = U \gg t$)

$$\hat{H}_{\text{eff}} = \sum_j \left(U + \frac{2t^2}{U} \right) |B_j\rangle \langle B_j| + \frac{t^2}{U} \sum_j \left(|B_j\rangle \langle B_{j-1}| + |B_j\rangle \langle B_{j+1}| \right)$$

Density of States: Effective Hamiltonian



Doublon Local Density of States ($U = 15$)

$$\text{Width} = \frac{4t^2}{U}$$

Summary

- **Mapping:** 1D Spin Magnons correspond exactly to interacting fermions.
- **Bound States:** Exact Green's function reveals formation of Doublons strictly when $|U| > 2t$.
- **Emergent Dynamics:** Feshbach projection proves the Doublon behaves as a heavy, composite quasiparticle.

Thank You

1. The Non-Local String Mapping

$$c_j^\dagger = \exp\left(i\pi \sum_{i=1}^{j-1} \sigma_i^+ \sigma_i^-\right) \sigma_j^+$$

$$c_j = \exp\left(-i\pi \sum_{i=1}^{j-1} \sigma_i^+ \sigma_i^-\right) \sigma_j^-$$

2. Longitudinal (Z) Coupling to Density

$$n_i = c_i^\dagger c_i = \sigma_i^+ \sigma_i^- = \frac{1 + \sigma_i^z}{2} \quad \Longrightarrow \quad J_z \sigma_i^z \sigma_{i+1}^z \longrightarrow 4 J_z n_i n_{i+1} + \text{const.}$$

3. Transverse (XY) Exchange to Hopping

$$\sigma_i^+ \sigma_{i+1}^- = c_i^\dagger (-\sigma_i^z) c_{i+1} = c_i^\dagger (1 - 2 c_i^\dagger c_i) c_{i+1} = c_i^\dagger c_{i+1}$$

Additional Slides: Feshbach Projection Formalism

$$\hat{H} = \begin{pmatrix} H_{PP} & H_{PQ} \\ H_{QP} & H_{QQ} \end{pmatrix} \quad |\psi\rangle = \begin{pmatrix} |\psi_P\rangle \\ |\psi_Q\rangle \end{pmatrix}$$

$$H_{PP}|\psi_P\rangle + H_{PQ}|\psi_Q\rangle = \omega|\psi_P\rangle$$

$$H_{QP}|\psi_P\rangle + H_{QQ}|\psi_Q\rangle = \omega|\psi_Q\rangle$$

$$(\omega - H_{QQ})|\psi_Q\rangle = H_{QP}|\psi_P\rangle$$

$$|\psi_Q\rangle = (\omega - H_{QQ})^{-1} H_{QP}|\psi_P\rangle$$

$$\implies \underbrace{\left[H_{PP} + H_{PQ}(\omega - H_{QQ})^{-1} H_{QP}\right]}_{\hat{H}_{\text{eff}}(\omega)} |\psi_P\rangle = \omega|\psi_P\rangle$$

Additional Slides: Green's Function Recurrence Relation

1. Definition

$$G(m,n;k,\omega) = \langle k,m| \left(\omega + i\eta - \hat{H}\right)^{-1} |k,n\rangle \equiv a_{m,n}$$

2. Action of the Hamiltonian Applying $\hat{H}$ yields a kinetic relative-hopping term $f(k)$ and an on-site interaction U:

$$H|k,n\rangle = -2t\cos\left(\frac{ka}{2}\right)\left(|k,n-1\rangle + |k,n+1\rangle\right) + U\delta_{n,1}|k,n\rangle$$

3. The Difference Equation Using $\mathbb{I} = \hat{G}(\omega)(\omega + i\eta - \hat{H})$, we obtain a second-order linear recurrence relation:

General Recurrence Relation

$$\delta_{m,n} = (\omega + i\eta - U\delta_{n,1})a_{m,n} + f(k)\left(a_{m,n-1} + a_{m,n+1}\right)$$

Additional Slides: Solving for $G(1,1;k,\omega)$

1. System of Equations (Fixing $m=1$)

$$\begin{aligned} n=1: &\quad (\omega+i\eta-U)a_1+f(k)a_2=1 \\ n\geq 2: &\quad (\omega+i\eta)a_n+f(k)(a_{n-1}+a_{n+1})=0 \end{aligned}$$

2. Ansatz & Characteristic Equation Assume a localized relative wavefunction $a_n=\alpha z^n$ (requiring $|z|<1$ for normalizability).

$$(\omega+i\eta)+f(k)\left(z+\frac{1}{z}\right)=0$$

3. Exact Solution Matching the root z to the boundary condition at $n=1$ yields the closed form:

Nearest-Neighbor Green's Function

$$G(1,1;k,\omega)=\frac{1}{\omega+i\eta-U+f(k)z}$$